

Effects of grapefruit, grapefruit juice and water preloads on energy balance, weight loss, body composition, and cardiometabolic risk in free-living obese adults



Background Reducing dietary energy density has proven to be an effective strategy to reduce energy intakes and promote weight control. This effect appears most robust when a low energy dense preload is consumed before meals. Yet, much discussion continues regarding the optimal form of a preload. The purpose of the present study was to compare effects of a solid (grapefruit), liquid (grapefruit juice) and water preload consumed prior to breakfast, lunch and dinner in the context of caloric restriction. **Methods** Eighty-five obese adults (BMI 30-39.9) were randomly assigned to (127 g) grapefruit (GF), grapefruit juice (GFJ) or water preload for 12 weeks after completing a 2-week caloric restriction phase. Preloads were matched for weight, calories, water content, and energy density. Weekly measures included blood pressure, weight, anthropometry and 24-hour dietary intakes. Resting energy expenditure, body composition, physical performance and cardiometabolic risk biomarkers were assessed. **Results** The total amount (grams) of food consumed did not change over time. Yet, after preloads were combined with caloric restriction, average dietary energy density and total energy intakes decreased by 20-29% from baseline values. Subjects experienced 7.1% weight loss overall, with significant decreases in percentage body, trunk, android and gynoid fat, as well as waist circumferences (-4.5 cm). However, differences were not statistically significant among groups. Nevertheless, the amount and direction of change in serum HDL-cholesterol levels in GF (+6.2%) and GFJ (+8.2%) preload groups was significantly greater than water preload group (-3.7%). **Conclusions** These data indicate that incorporating consumption of a low energy dense dietary preload in a caloric restricted diet is a highly effective weight loss strategy. But, the form of the preload did not have differential effects on energy balance, weight loss or body composition. It is notable that subjects in GF and GFJ preload groups experienced significantly greater benefits in lipid profiles. Trial registration ClinicalTrials.gov NCT00581074